GEORGETOWN UNIVERSITY · DSAN 6725 FINAL PROJECT

Agentic AI Nutritionist

A multi-agent meal-planning system that combines persistent user memory with grounded recipe retrieval and a deterministic audit layer.

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IN BRIEF A multi-agent system that produces **personalized, nutritionally targeted meal plans** through conversation — grounded in a 751-recipe corpus, with a deterministic audit layer outside the LLM.
Perfect on preliminary safety and structure tests; corpus and latency are the remaining limits.

1 The Problem

motivation

Manual trackers like MyFitnessPal place the data-entry burden on the user, while purely generative LLMs struggle to produce reliable nutritional values.

695 *kcal* **Five leading LLMs** systematically *underestimated* daily energy vs. registered-dietitian plans.
Bilen et al., 2026 · n = 5 models

We explore a middle path: an **agentic architecture** that combines persistent memory with grounded retrieval — and **isolates safety from LLM inference**.

Related work · *Angara 2017, Abbas 2026, Guo 2025, Bilen 2026, Khamesian 2024, Papastratis 2024.*

2 The Corpus

budget bytes · 751

Scraped from **Budget Bytes**. Category labels route recipes to the right meal slot.

751	$1.22	35 *min*	213 *kcal*
RECIPES	MEDIAN COST / SERVING	MEDIAN PREP + COOK	MEAN KCAL / SERVING

CAVEAT Nutrition values were *LLM-estimated* — not USDA. Production would replace with a lab-validated dataset.

3 The Stack

why these

- `claude-sonnet-4.6` — ORCHESTRATOR + INTAKE
- `claude-haiku-4.5` — PLANNER · STRUCTURED
- `strands-agent-sdk` — AGENT FRAMEWORK
- `mem0` — PERSISTENT MEMORY
- `sqlite` — PERSONAL COOKBOOK
- `streamlit` — FRONTEND

Chosen to keep the *what the model is doing* legible — Strands over LangChain, SQLite over vector-store.

4 System Architecture

figure 1

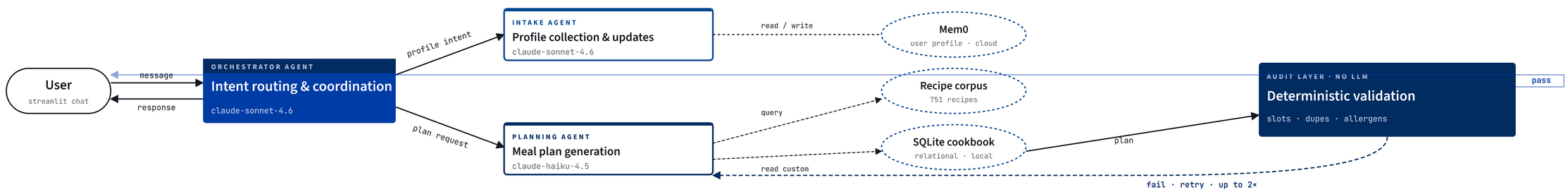

Figure 1. *The* **Orchestrator** *(Sonnet 4.6) routes intent to an* **Intake Agent** *(Sonnet 4.6, reads/writes Mem0) or the* **Planning Agent** *(Haiku 4.5, queries the recipe corpus + SQLite cookbook). The* **Audit Layer** *(no LLM) validates every plan for allergens, duplicates, and required slots — looping the Planner up to two times before falling back.*

5 The Safety Loop

no LLM in the validator

01 **Request** — orchestrator
→ 02 **Compute targets** — mifflin-st jeor
→ 03 **Query corpus** — filter · score
→ 04 **Assemble** — 1 or N days
→ *Audit?* — slots · allergens
→ ✓ PASS **Present** — to user

RETRY · 1: **Widen time**
RETRY · 2: **Widen ingredient count**
FALLBACK: **Best plan + notes**

6 How We Tested

braintrust · no LLM judge

29 CASES · 5 CATEGORIES

Meal Planning	12
Dietary Restriction	6
Profile Update	4
Out of Scope	4
Shopping List	3
Total	**29**

8 DETERMINISTIC SCORERS

- `ToolSelection`
- `NoError`
- `PlanCompleteness`
- `AllergenSafety`
- `Latency`
- `ScopeAwareness`
- `CalorieTarget`
- `ShoppingList`

Binary or banded numeric. CalorieTarget band: ±15% of Mifflin–St Jeor.

7 Results

figure 2 · table 1

FIG. 2 · AVG. SCORE BY SCORER

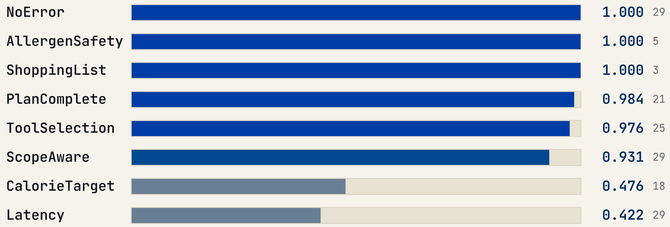

TABLE 1 · PER-CATEGORY (APP. B)

CATEGORY	TOOL	PLAN	CAL	ALLG	SCOPE	LAT
Meal Planning	0.983	0.986	0.518	–	1.000	0.208
Dietary Restr.	0.933	0.972	0.392	1.000	1.000	0.417
Shopping List	1.000	1.000	–	–	1.000	0.167
Profile Update	1.000	–	–	–	1.000	0.750
Out of Scope	–	–	–	–	0.500	0.938

Fig. 2 & Table 1. *Safety-critical and structural scorers are saturated. CalorieTarget (corpus-bound) and Latency (sequential inference) are unpacked in §8. Bar n = applicable cases; table bodies dashed where scorer does not apply.*

8 What It Means

findings

01 Grounding eliminates hallucination — by construction.
The system can't invent meals. The audit is string-matching.

02 Separated agents made debugging tractable.
Narrow prompts, predictable routing, isolated failures.

03 CalorieTarget (0.476) is a *corpus* problem.
Budget Bytes skews light (213 kcal/srv). USDA corpus closes most of the gap.

04 Latency (0.422) is an *infra* problem.
2–3 sequential API calls per turn. Parallelize + cache to fix.

9 Conclusion

& references

TAKEAWAY

Separating conversation, retrieval, and validation across specialized agents allowed the system to stay conversational while keeping nutritional accuracy and dietary constraints reliable.

Next: USDA corpus · auth · grocery-MCP · temporal profile.

REPO · DEMO

REFERENCES

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[2] Angara, P. *et al.* (2017). Foodie Fooderson: A conversational agent for the smart kitchen. *CASCON 2017*.
[3] Bilen, A. B. *et al.* (2026). AI diet plans underestimate nutrient intake vs. dietitians. *Frontiers in Nutrition*, 13, 1765598.
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[5] Khamesian, S. *et al.* (2024). NutriGen: Personalized meal-plan generator leveraging LLMs. *arXiv*.
[6] Papastratis, I. *et al.* (2024). AI nutrition recommendation using a generative model and ChatGPT. *Sci. Reports*, 14, 14620.

Colophon. *Poster design iterated with Claude (Anthropic) via its design-tool workspace — layout, typography, and SVG figures were generated through conversation with the model.*